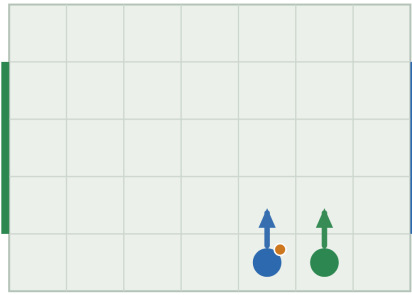


Player positions and the 4×4 Q matrix

At the research meeting the professor drew a stage game by hand: a small grid of cells, each marked, with arrows chasing each other around the grid to show that no cell is a stable outcome, at players positioned $(0, 0)$ and $(1, 1)$. This is that sketch, generated from the exact solver instead of drawn free-hand, paired with a picture of where the two players actually are, and the exact, complete 4×4 $\{U, D, L, R\}$ Q matrix underneath every case — always the full grid, never reduced, per the professor's own correction that the matrix should stay 4×4 and that the matrix itself, not an entropy number, is the actual output.

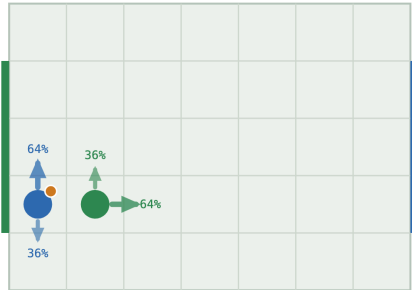
Twelve cases. Across the whole project there are exactly four *canonical* equilibrium-support shapes once every stage game is reoriented so rows are always the carrier's actions — $(2, 2)$: 68 states, $(3, 3)$: 4, $(2, 1)$: 14, $(3, 2)$: 8. **The defender's support is never larger than the carrier's**, in any of the 94 mixed states on the canonical board. This page shows one example of all four shapes, plus every distinct mechanism studied in the project that can force a mix: a stochastic move order, this project's own tackle rule, a dense reward with zero transition noise, and movement slip on a goal shape that is otherwise always pure.

A pure state -- one safe cell, nothing to guess



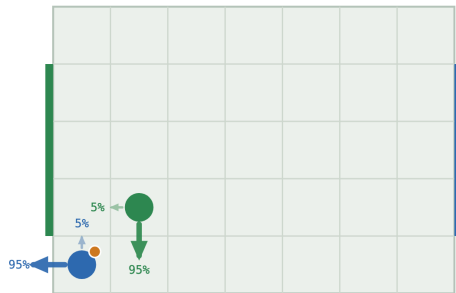
pure equilibrium · $V = +0.234$

The typical mix -- which goal row to head for



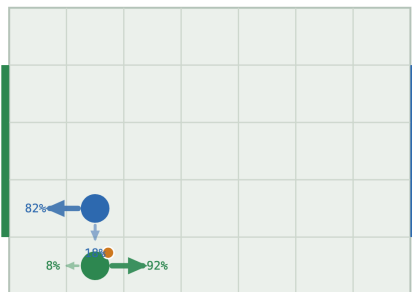
mixed equilibrium · $V = +0.094$

The corner duel -- (0,0) and (1,1)



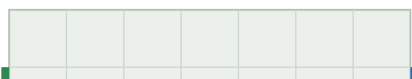
mixed equilibrium · $V = +0.078$

A clean L/R indifference example

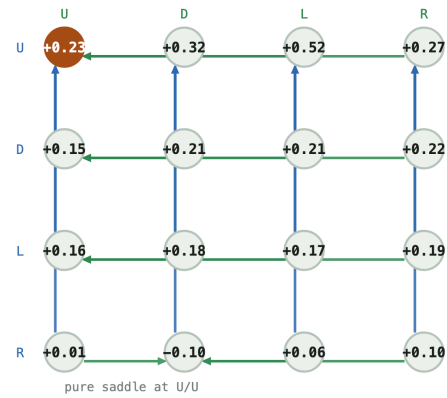


mixed equilibrium · $V = -0.206$

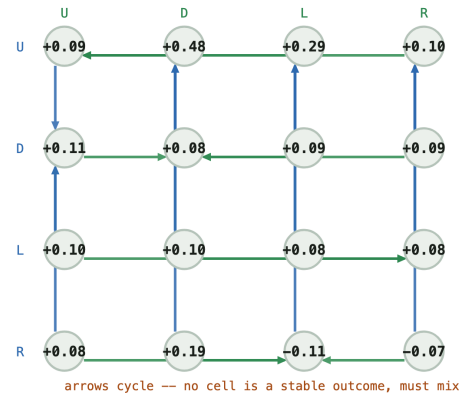
A genuine 3-action mix, not a 2-cycle



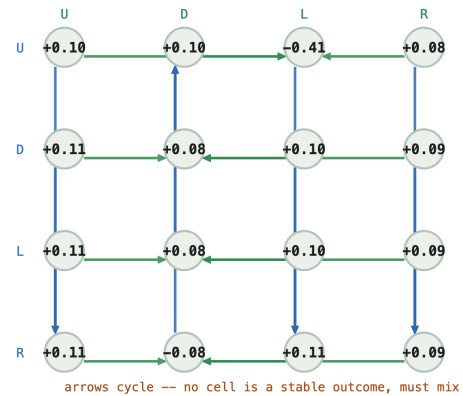
Q matrix -- (4, 0, 5, 0, 0)



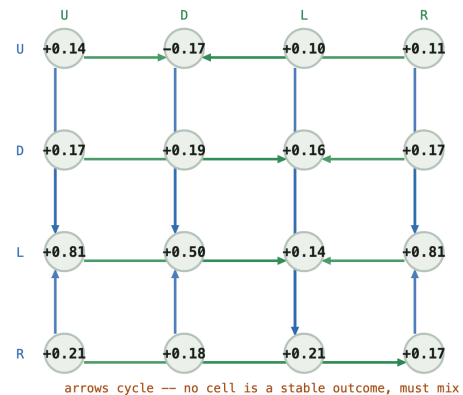
Q matrix -- (0, 1, 1, 1, 0)



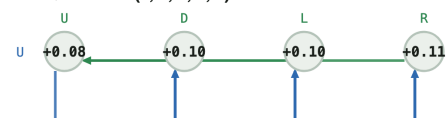
Q matrix -- (0, 0, 1, 1, 0)

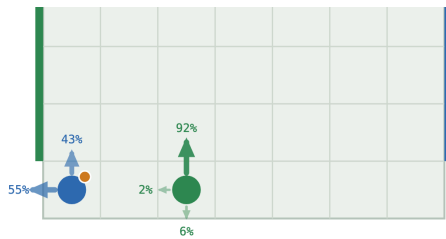


Q matrix -- (1, 1, 1, 0, 1)



Q matrix -- (0, 0, 2, 0, 0)





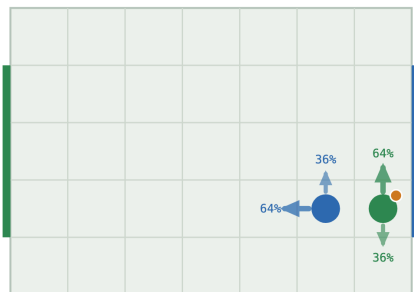
mixed equilibrium · $V = +0.085$

A near-pure hedge -- where rounding would lie



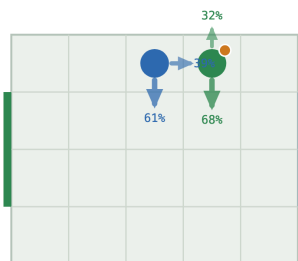
mixed equilibrium · $V = -0.168$

The typical mix, mirrored -- same shape, value negated



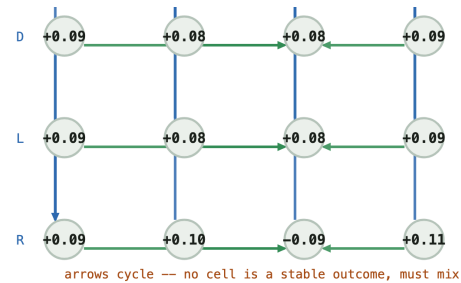
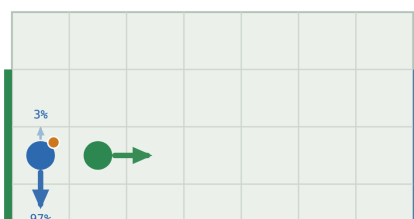
mixed equilibrium · $V = -0.094$

This project's own tackle rule -- a different mechanism, same duel

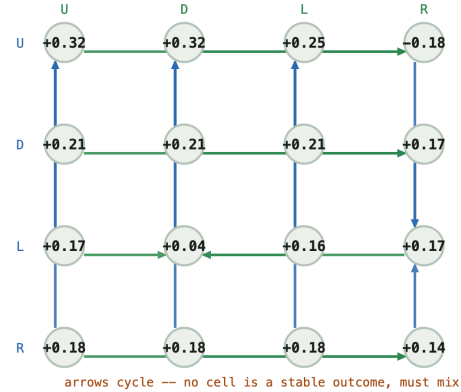


mixed equilibrium · $V = +0.013$

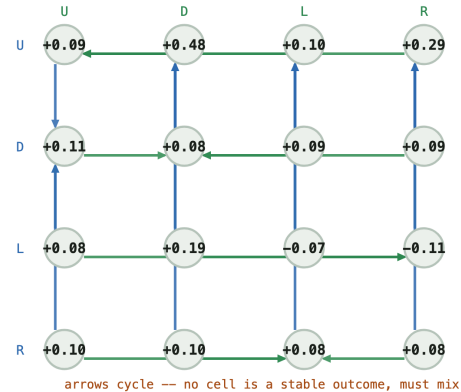
The asymmetric mix -- the carrier still guesses, the defender doesn't



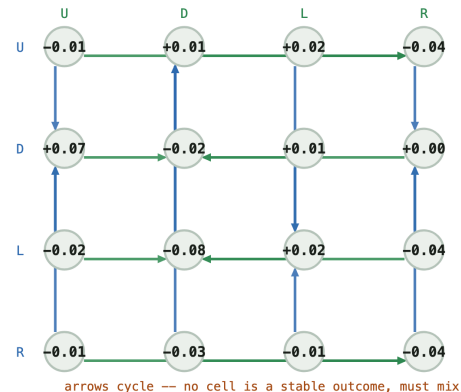
Q matrix -- (1, 1, 2, 0, 1)



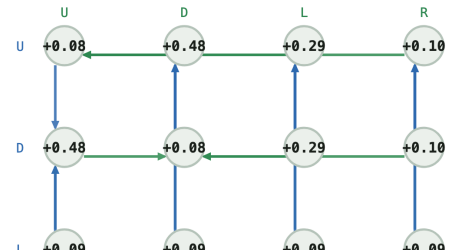
Q matrix -- (5, 1, 6, 1, 1)

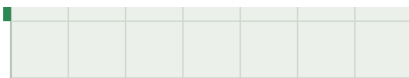


Q matrix -- (2, 3, 3, 3, 1)



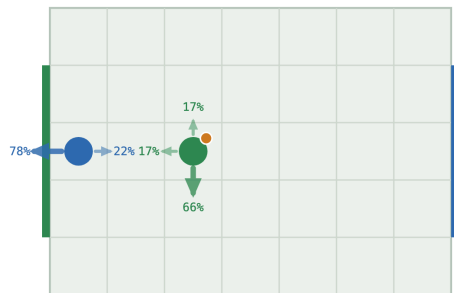
Q matrix -- (0, 2, 1, 2, 0)





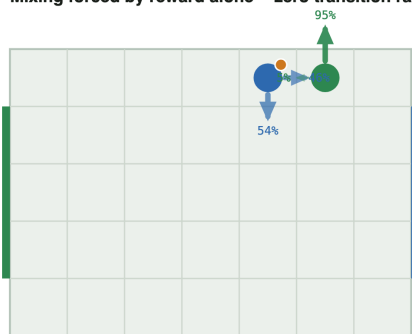
mixed equilibrium · $V = +0.095$

The three-lane mix -- support (3,2), the deepest gap on this page



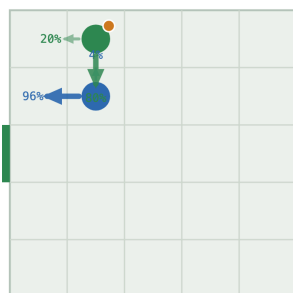
mixed equilibrium · $V = -0.282$

Mixing forced by reward alone -- zero transition randomness

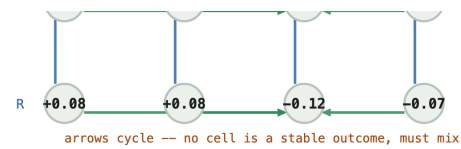


mixed equilibrium · $V = +0.129$

Movement slip -- a single-cell goal, always pure until now

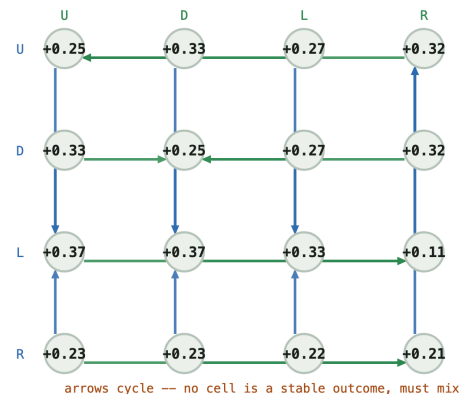


mixed equilibrium · $V = -0.053$



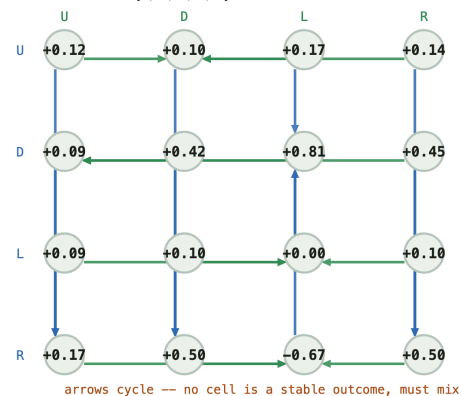
arrows cycle -- no cell is a stable outcome, must mix

Q matrix -- (0, 2, 2, 1)



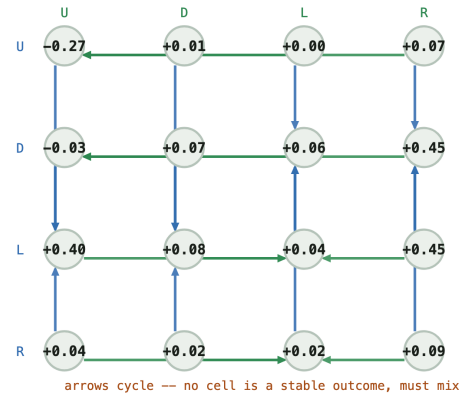
arrows cycle -- no cell is a stable outcome, must mix

Q matrix -- (4, 4, 5, 4, 0)



arrows cycle -- no cell is a stable outcome, must mix

Q matrix -- (1, 3, 1, 4, 1)



arrows cycle -- no cell is a stable outcome, must mix

All twelve cases: board position (left) and the full 4x4 Q matrix redrawn as a node-and-arrow graph (right). A lit node with no outgoing arrow is a pure saddle; a closed loop of arrows means no cell is safe.

Reading the graph

Each node is one cell of the complete 4x4 Q matrix (row = the carrier's action, col = the defender's action, always

reoriented so the printed payoff is the carrier's, regardless of which player id actually carries the ball). A blue arrow points from a cell to the cell in the same column with the higher payoff — the carrier's own reason to switch rows. A green arrow points from a cell to the cell in the same row with the lower payoff — the defender's reason to switch columns, since the defender minimises. Every case shows all 16 cells, so the shapes are directly comparable.

Case 1 (4, 0, 5, 0, 0)

A pure state, for contrast. Saddle at U/U. The carrier plays U always; the defender answers U always; neither has any reason to deviate.

	U	D	L	R
U	0.234	0.317	0.523	0.272
D	0.151	0.211	0.211	0.222
L	0.161	0.182	0.172	0.190
R	0.012	-0.095	0.058	0.098

Case 2 (0, 1, 1, 1, 0)

The typical mix. Gap 0.0136. The carrier mixes U 63.5% / D 36.5%; the defender answers U 36.5% / R 63.5%. This is the common shape: the carrier is deciding which goal row to head for, and the defender is guessing which one. **90 of the 94 mixed states on the canonical board cross a vertical pair exactly like this one.**

	U	D	L	R
U	0.086	0.478	0.286	0.099
D	0.109	0.076	0.085	0.086
L	0.103	0.103	0.085	0.084
R	0.082	0.187	-0.108	-0.071

Case 3

(0, 0, 1, 1, 0)

The corner duel. The board position sketched at the meeting: the carrier at (0, 0) — the back corner, right against its own goal — and the defender diagonally adjacent at (1, 1). This is the closest match, coordinate for coordinate, to the meeting's own sketch. Distinct from Case 2 (which crosses U/D), this one crosses U/L for the carrier and D/L for the defender: pinned in the corner, the carrier's two live escapes are straight up the sideline (U) or across along the back line (L), and the defender is guessing which. **Gap 0.0121; the mix is lopsided but genuine** (U 4.6% / L 95.4% for the carrier, D 94.9% / L 5.1% for the defender) — the corner leaves little room, but not zero.

	U	D	L	R
U	0.103	0.103	-0.408	0.084
D	0.109	0.076	0.101	0.086
L	0.109	0.076	0.101	0.086
R	0.112	-0.075	0.112	0.088

Case 4 (1, 1, 1, 0, 1)

A clean L/R indifference example.

"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."

Gap 0.0445. The carrier's entire live option set is exactly $\{L, R\}$ — no vertical option survives at all, which is what makes this state (rather than Case 3, which crosses U/L) the cleanest match to the meeting's *verbal* description: two actions genuinely equally good against the defender's own mix (D 18% / L 82%). **This is the rarer shape overall: only 4 of 94 mixed states cross a purely horizontal pair instead of a vertical one.**

	U	D	L	R
U	0.139	-0.173	0.099	0.110
D	0.171	0.185	0.157	0.171
L	0.810	0.500	0.141	0.810
R	0.211	0.181	0.211	0.167

Case 5 (0, 0, 2, 0, 0)

A genuine 3-action mix. Gap 0.0047, entropy 1.109 bits — the richest mix on this page. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two. Requested explicitly at the meeting: **the matrix itself, not entropy, is the point.**

	U	D	L	R
U	0.084	0.095	0.103	0.110
D	0.086	0.076	0.076	0.086
L	0.086	0.076	0.076	0.086
R	0.088	0.095	-0.088	0.110

Case 6 (1, 1, 2, 0, 1)

A near-pure hedge, where rounding would lie. Gap 0.0043, entropy only 0.169 bits — the carrier plays R 97.5%, D 2.5%. Reading 0.975 as 1.0 would be the rounding mistake the meeting flagged: the gap is real and certified, twelve times smaller than a 0.1 rounding grid would resolve.

	U	D	L	R
U	0.317	0.317	0.247	-0.178
D	0.211	0.211	0.211	0.167
L	0.171	0.045	0.157	0.171
R	0.182	0.182	0.182	0.135

Case 7 (5, 1, 6, 1, 1)

The typical mix, mirrored. The board's left-right / role-swap symmetry maps Case 2 exactly onto this state. The values negate to machine precision:

V(case 2) = +0.094235
V(mirror) = -0.094235
sum = -1.39e-17 (zero, to floating-point noise)

Direct evidence that Case 2's shape is not a one-off: the identical mix reappears, exactly mirrored, wherever the same relative configuration recurs.

	U	D	L	R
U	0.086	0.478	0.099	0.286
D	0.109	0.076	0.086	0.085
L	0.082	0.187	-0.071	-0.108
R	0.103	0.103	0.084	0.085

Case 8

(2, 3, 3, 3, 1), 5×4 board

This project's own tackle rule. Every other case here gets its coin flip from Littman's random move order. This one is solved under the tackle rule (defender commits to a challenge that wins the ball with fixed probability or bounces off) instead — a mechanism with nothing to do with move order at all. **It produces the same kind of duel** (gap 0.0223): the carrier mixes D 60.7% / R 39.3%, the defender mixes U 32.1% / D 67.9%. Different cause, same structure.

	U	D	L	R
U	-0.012	0.005	0.016	-0.042
D	0.072	-0.022	0.009	0.000
L	-0.019	-0.080	0.019	-0.042
R	-0.007	-0.026	-0.006	-0.039

Case 9

(0, 2, 1, 2, 0)

The asymmetric mix. Support (2, 1) — gap 0.0095. The odd one out: the carrier's equilibrium still mixes two actions (U 2.6% / D 97.4%), but the defender's equilibrium is a single fixed move (R, 100%). Every other case on this page is a symmetric duel — both players genuinely guessing. Here only one side is.

	U	D	L	R
U	0.085	0.478	0.291	0.095
D	0.478	0.085	0.291	0.095
L	0.093	0.093	0.086	0.093
R	0.082	0.082	-0.122	-0.072

Why the carrier still needs two actions against a defender who never varies: against the defender's pure R, column R reads U 0.095, D 0.095, L 0.093, R -0.072 — **U and D are exactly tied**, both strictly better than L or R. Nothing in the payoffs favours one over the other, so any split of U/D is an equally valid best reply; the solver's LP reports one particular split (2.6% / 97.4%), not a probability forced by the game the way Case 4's 18% / 82% is. This is the meeting's LP-edge-case question made concrete: **a tie between two rows can look identical, in the printed policy, to a genuine forced mix, but it is a different phenomenon** — no best-reply cycle is involved, and either pure U or pure D alone would also be a valid equilibrium reply to the defender's fixed R.

Case 10 (0, 2, 2, 2, 1)

The three-lane mix. Support (3, 2) — gap 0.0699, **the deepest gap of any state on this page** (fifteen times deeper than the shallowest hedge in Case 6). This is the fourth and last canonical support shape: the carrier genuinely needs three live actions (U 16.9% / D 66% / L 17.1%), while the defender only ever needs two (L 77.9% / R 22.1%). Combined with Case 9, this settles a structural question the certificates make checkable rather than assumed: **the defender never needs strictly more actions than the carrier**, across all 94 mixed states on the canonical board.

	U	D	L	R
U	0.247	0.330	0.272	0.317
D	0.330	0.247	0.272	0.317
L	0.366	0.366	0.330	0.110
R	0.228	0.228	0.220	0.205

Case 11 (4, 4, 5, 4, 0), deterministic

Mixing forced by reward alone. Every case so far gets its coin flip from a stochastic transition — the random move order, or the tackle rule's own coin. This one has **no transition stochasticity at all**: the mix comes entirely from `scoring="territory"`, a dense per-step reward for the ball in the opponent's final third, layered on top of the ordinary win/loss score. Gap 0.0653, entropy 0.996 bits — close to a fair coin, forced purely by coupling the reward to both players' actions.

	U	D	L	R
U	0.116	0.105	0.170	0.136
D	0.094	0.415	0.815	0.450
L	0.094	0.105	0.000	0.105
R	0.170	0.500	-0.671	0.500

Case 12 (1, 3, 1, 4, 1)

Movement slip on a single-cell goal. The sharpest possible contrast with Case 1. A single-cell goal under deterministic move order is provably pure at every one of its states — that is the whole goal-width result. Add `slip=0.15` (each player independently takes a uniform-random move instead of its chosen one, 15% of the time, regardless of position) to that exact board, and **52 mixed states appear where there were 0**. Gap 0.0172 — a genuine, certified mix, not noise. The goal-width switch is a property of Littman's move-order coin specifically; a coin that lands on every square regardless of the goal forces mixing everywhere, independent of goal width.

	U	D	L	R
U	-0.273	0.006	0.001	0.069
D	-0.031	0.069	0.057	0.450
L	0.396	0.083	0.039	0.450
R	0.039	0.020	0.015	0.092

The mathematics behind all twelve

A mixed equilibrium is exactly the strategy pair where every action in a player's support earns the **same expected payoff** against the opponent's own mix — that equality is what "indifferent" means in Case 4, and it is also why the best-response graph cycles with no resting point: if one action in the support paid strictly more, that player would raise its weight on it, which is precisely the condition an equilibrium rules out. A best-reply cycle and mutual indifference are the same fact seen from two sides, not two different explanations. Case 9 shows the boundary of that story: indifference alone (a tie against a fixed opponent) can produce the same *symptom* — a fractional policy — without a cycle anywhere in the matrix.

Reproduce with `python scripts/positions.py` (make `positions`) — writes `figures/gallery/positions.svg` and prints every matrix and policy shown here. Full project: github.com/Charith-Reddy-Pareddy/soccer-markov-nash.